

Project Brief Template

For students proposing their own project

PURPOSE

Complete this brief and submit it to the mentors for approval before full implementation. Define the business problem and propose a technically realistic ML solution without writing a step-by-step implementation recipe.

1. Project Identification

Decision / Question	Student Response
Student full name	Shakhnoza Abdusalomova
Project title	Automated Fabric Defect Classification Using Deep Learning for Textile Quality Inspection
Industry / field	Textile Manufacturing, Quality Control, Computer Vision
Intended client / user / stakeholder	Textile manufacturers, quality inspection teams, garment production companies, and fabric quality control inspectors
One-sentence project summary	An AI-based computer vision system that automatically detects and classifies defects in fabric images to assist quality inspection during textile manufacturing.

2. Client / User Background

Describe the organization, user or context in which the problem exists. Explain only what is necessary to understand the project.

Textile manufacturers inspect fabric before it is used in clothing production. This inspection is often performed manually, which can be time-consuming, inconsistent, and affected by human fatigue. Detecting defects early helps reduce waste, improve product quality, and lower production costs.

3. Business Problem

What is currently difficult, slow, expensive, inconsistent or impossible? Why does the problem matter?

Manual fabric inspection is repetitive and may fail to identify small or subtle defects consistently. Missed defects can lead to lower product quality, customer complaints, and financial losses. An automated computer vision system can assist inspectors by providing

faster and more consistent defect detection.

4. Requested Solution

Describe the outcome you want the ML/AI system to provide. Do not describe the exact algorithm yet.

Develop a machine learning system that analyzes fabric images and predicts whether the fabric contains defects. Depending on the selected dataset, the system may also classify the type of defect. The system is intended to support quality inspectors rather than replace human decision-making.

5. Available / Expected Information

What types of data could realistically exist for this problem? Do not assume access to private data you cannot legally use.

Data sources

Fabric images

Defect labels

Defect categories (if available)

Public textile defect dataset

Image annotations provided by the dataset

6. Data & Problem Discovery

Decision / Question	Student Response
Selected dataset and source	Multi-Class Fabric Defect Detection Dataset
What does one record / sample represent?	One sample represents a single high-resolution image of a fabric surface captured during textile production. Each image belongs to one of the nine defect categories.
Proposed target or ML objective	Predict the fabric defect category from an input fabric image.
Key information available at prediction / inference time	A colour image (1280 × 720 pixels) of a fabric surface captured during the manufacturing process.

Main data quality issues	Imbalanced class distribution (some defect types occur much less frequently than others). Variations in fabric textures and colours. Differences in defect size and appearance. Some defect classes may be visually similar, making classification more challenging.
Potential leakage risks	Images from the same production batch or fabric roll could appear in both the training and testing sets, potentially leading to overly optimistic evaluation results. Careful dataset splitting will be used to minimise this risk.
Privacy / fairness / licensing concerns	The dataset contains only images of fabric and does not include any personal or sensitive information. Therefore, there are no significant privacy concerns. Model performance will be evaluated across all defect classes to ensure that minority classes are not systematically overlooked.

7. Technical Proposal

Decision / Question	Student Response
ML problem formulation	Multi-class image classification.
Proposed baseline	CNN
Main modeling approach(es) to investigate	ResNet18
Data splitting / validation strategy	Train, validation, and test split with performance evaluated on unseen images.
Primary evaluation metric(s) and why	Macro F1, recall of rare defects
Expected inference input	A single RGB fabric image
Expected inference output	Predicted defect category with confidence score.
Main technical risks / assumptions	Imbalanced classes Limited images for some defect categories Visually similar defects Overfitting

8. Functional Requirements

Write 4–7 outcome-focused requirements describing what the finished system must be able to do.

1. Accept a fabric image as input.
2. Preprocess the image before prediction.
3. Classify the fabric into one of the defect categories.
4. Display the predicted class and confidence score.
5. Evaluate the trained model using standard classification metrics.

9. Expected Deliverables

- Trained deep learning model.
- Jupyter notebooks.
- Source code.
- Performance evaluation report.
- Final presentation.
- GitHub repository.

Note: General capstone requirements are defined in the official Capstone Evaluation Criteria and Helper documents.

10. Acceptance Criteria

Define what must be true for the project to be considered functionally complete.

The model successfully classifies unseen fabric images.
Evaluation metrics are reported.
The project documentation is complete.
The code is reproducible.
The GitHub repository is organised and well documented.

11. Constraints — In Scope / Out of Scope

IN SCOPE	OUT OF SCOPE
Dataset exploration	Real-time factory deployment
Data preprocessing	Industrial camera integration
Transfer learning	Mobile application
Model training	Robotic quality inspection
Model evaluation	
Prediction on new image	

12. Questions You Must Resolve

List the key engineering/product questions that must be answered before or during implementation.

Which pretrained model provides the best performance?
How should class imbalance be handled?
What image preprocessing techniques improve accuracy?
Which evaluation metric best reflects model performance?

13. Optional Directions

List optional extensions only after the core scope is clear. These must not be required for the project to be considered complete.

- ____Real-time defect classification using a live camera.
 - Defect localisation using object detection.
 - Edge deployment on embedded devices.
 - Automatic quality control reporting.
 - Integration into textile production lines.

14. Mentor Review & Approval

This section is completed by the mentor after reviewing the proposed scope and technical direction.

Decision	☐ Approved ☐ Approved with revisions ☐ Revision required
Required revisions / comments	
Approved scope / special conditions	
Mentor name	
Date	